Statistical Mechanics

Bohaz

December 1st 2025

Introduction

This note offers an alternative perspective to the Year 2 lectures in the MChem course and presents it succinctly.

Quantum mechanics describes the microscopic properties of atoms and molecules, while Thermodynamics describes the properties of bulk matter. The two are bridged by statistical mechanics, which regards thermodynamics as a logical consequence of our ignorance of the macroscopic system due to our incomplete knowledge of the positions and momenta of every particle, or the sets of quantum numbers required to specify the states.

I - Entropy

We typically resort to the use of probability when we do not have complete knowledge of a given system. To quantify our ignorance or missing information, we introduce a measure called **Entropy**, which assigns a number s to a probability distribution $\{p_i\}$. We will rewrite $s(\{p_i\})$ as $s(p_i)$ to avoid clutter.

Logic dictates that entropy s must satisfy three properties:

1. Continuous: a small variation in the distribution $\{p_i\}$ should result in a small variation of s
2. Monotonic: a uniform distribution should give the largest s, more possible outcomes results in greater s.
3. Linear: consistent with a case that can be broken down into smaller cases.

From the second statement, we define

$$s_n \equiv s\big(\frac{1}{n}\big) \tag{I.1}$$

which is the maximum value of s, for possible outcomes n and satisfy $s_{n+1} > s_n$.

An example that illustrates the requirement of the third statement is

$$s(p_1, p_2, p_3) = s(p_1, p_{23}) + p_{23} s\Big(\frac{p_2}{p_{23}}, \frac{p_3}{p_{23}}\Big) \tag{I.2}$$

where $p_{23} = p_2 + p_3$, and the probabilities in the bracket of the second term of Eq. (I.2) are the conditional probabilities given that either 2 or 3 has occurred. In other words, arbitrarily grouping the outcomes together does not mean we become any less ignorant of the system; rather, with the ignorance within each group is added and weighed by the probability of being found in the group.

Consider a system with N equally likely outcomes, the outcomes form n group, with n_i outcomes in the ith group. Therefore the probability of an outcome found to be in the ith group is $p_i = n_i/N$, and any possibility in i^{th} group given that we are already in it is $1/n_i$.

Therefore we have,

$$s\Big(\frac{1}{N}\Big) = s(p_i) + \sum_i^n p_i s\Big(\frac{1}{n_i}\Big) \tag{I.3}$$

using Eq. (I.1) yields

$$s_N = s(p_i) + \sum_i^n p_i s_{n_i} \tag{I.4}$$

If we let $n_i = m$, then $N = nm$, thus

$$s_{nm} = s_n + s_m \tag{I.5}$$

thus we deduce

$$s_n = \ln n \tag{I.6}$$

substitute into Eq. (I.4) yields

$$\ln N = s(p_i) + \sum_i^n p_i \ln n_i \tag{I.7}$$

after a slight rearangement, we arrive at

$$s \equiv -\sum_i p_i \ln p_i \tag{I.8}$$

II - Temperature

The expectation value of energy is defined as

$$\langle H \rangle \equiv \sum_i p_i E_i \tag{II.1}$$

where E_i are defined from the TISE $H|i\rangle = E_i|i\rangle$, it is worth noting that E_i is a function of volume in the sense of the boundary condition of the wavefunctions that solves TISE.

We shall now determine the p_i's.

We argue that the only rational way to assign the probabilities of being in a stationary state is that they should be consistent with macroscopic measurements and that it maximises entropy. We should know just enough, but we have no reason to know more.

The maximisation under constraint Eq. (II.1) and $\sum_i p_i = 1$ can be done using the method of Lagrange multiplier, by defining

$$\Phi(\{p_i\}, \alpha, \beta) = -\sum_i p_i \ln p_i - \alpha\Big(\sum_i p_i - 1\Big) - \beta\Big(\sum_i p_i E_i - \langle H \rangle\Big) \tag{II.2}$$

differentiating and setting to zero

$$\frac{\partial \Phi(\{p_i\}, \alpha, \beta)}{\partial p_i} = 0 \tag{II.3}$$

yields the Boltzmann distribution

$$p_i = \frac{1}{Z} e^{-\beta E_i} \tag{II.4}$$

where the partition function is defined by normalisation

$$Z \equiv \sum_i e^{-\beta E_i} = e^{1+\alpha} \tag{II.5}$$

It is easy to show that, at constant volume,

$$\langle H \rangle = -\frac{\partial \ln Z}{\partial \beta} \tag{II.6}$$

substitute Eq. (II.4) into the $\ln p_i$ in Eq. (I.8) yields

$$s = -\sum_i p_i(\beta E_i - \ln Z) = \beta \langle H \rangle + \ln Z \tag{II.7}$$

taking the differential at constant volume yields

$$ds = \beta d\langle H \rangle + \langle H \rangle d\beta - \frac{\partial \ln Z}{\partial \beta} d\beta = \beta d\langle H \rangle \tag{II.8}$$

Hence

$$\beta = \Big(\frac{\partial s}{\partial \langle H \rangle}\Big)_V \tag{II.9}$$

We then recognise that both internal energy and thermodynamic entropy in statistical terms as

$$U \equiv \langle H \rangle \tag{II.10}$$

and

$$S \equiv k_B s \tag{II.11}$$

where k_B carries the unit of thermodynamic entropy.
We then define temperature

$$T \equiv \Big(\frac{\partial U}{\partial S}\Big)_V \tag{II.12}$$

comparison with Eq. (II.9) gives

$$\beta = \frac{1}{k_B T} \tag{II.13}$$

Consider the quantity defined by

$$A \equiv -k_B T \ln Z \tag{II.14}$$

substitute into Eq. (II.7) and rearrange yields

$$A = U - TS \tag{II.15}$$

which is the definition of Helmholtz free energy in classical thermodynamics.

III - Heat and Work

Taking the differential of internal energy yields

$$dU = \sum_i \big(E_i dp_i + p_i dE_i\big) \tag{III.1}$$

Taking the differential of entropy yields

$$dS = -k_B \sum_i (\ln p_i + 1) dp_i = -k_B \sum_i \ln p_i dp_i \tag{III.2}$$

then using Eq. (II.4) we obtain

$$TdS = \sum_i E_i dp_i \tag{III.3}$$

At constant volume $dE_i = 0$, therefore, no work is done on the system. However, $dp_i \neq 0$ as p_i depends on the temperature, thus only heat is transferred to the system.

More subtly, p_i is obtained by maximising entropy for a given thermodynamic state, thus it is the probability distribution for a system at equilibrium, therefore the heat is transferred reversibly and we have

$$dS = \frac{đq_{rev}}{T} \tag{III.4}$$

which is just the definition of entropy in classical thermodynamics.

We now define pressure

$$P = -\sum_i p_i \frac{dE_i}{dV} \tag{III.5}$$

multiply both sides by dV yields

$$-PdV = \sum_i p_i dE_i \tag{III.6}$$

More subtly, given that V is slowly varying so that the $\hat{H}$ of the system is slowly varying, a well known theorem called the 'Adiabatic theorem' from time-dependent perturbation theory states that the system will stay in whatever stationary state $|i\rangle$ it started in. Therefore, at constant temperature p_i remains constant and $dp_i = 0$. Therefore, no heat is transferred and entropy of the system does not change, this is precisely the meaning of reversible adiabatic process, hence the name of the theorem.

$$đw_{rev} = -PdV \tag{III.7}$$

substituting Eq. (III.3) and (III.5) into (III.1) yields the fundamental equation

$$dU = TdS - PdV \tag{III.8}$$

thus we can obtain another expression for pressure

$$p = -\Big(\frac{\partial U}{\partial V}\Big)_S \tag{III.9}$$

which is the definition of pressure in classical thermodynamics.

The other expression is just the definition of temperature Eq. (II.12).

This completes a brief overview of the important concepts in statistical mechanics. We have shown that thermodynamics naturally arises as a logical consequence when we admit that there is a lot we don't know about a macroscopic system to express the wavefunction of the entire system in terms of quantum mechanics. We therefore resort to the use of probability and maximise the entropy which accounts for our ignorance in a way that is consistent with what is observed macroscopically.